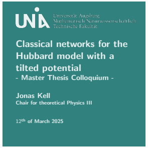
2025-03-12

# Master Colloquium Presentation




1. Thank you for coming
2. Title: Classical networks for the Hubbard model with a tilted potential
3. Quickly what this will be about

1. 
   - Introduction to the physical theory
   - In depth: Time evolution - why and how
   - Quickly: Sampling Strategy and simplifications
   - To give the title justice: Variational classical networks - what is is all about
   - End (if there is time or if questions lead into that direction): Implementation and experiments
   - Outlook

## Particle Types

### Introduction of the Physical Problem

- Bosonic operators
  - Commutation relations

$$\left[\hat{b}_l, \hat{b}_m\right] = 0$$
$$\left[\hat{b}_l^\dagger, \hat{b}_m^\dagger\right] = 0$$
$$\left[\hat{b}_l, \hat{b}_m^\dagger\right] = \delta_{l,m}$$

1. Wanna discuss, because during the writing of the thesis quite new insight in what in fact the differences of behavior are in between the particle types
2. First of all wanna point out how we have lattice operators, so fixed sites where there can be particles
3. Occupation number from 0 to infinity
4. Nice to handle because they commute

## Particle Types

### Introduction of the Physical Problem

- Bosonic operators
  - Commutation relations
- Fermionic operators
  - Anti-commutation relations

$$\left\{ \hat{c}_l, \hat{c}_m \right\} = 0$$
$$\left\{ \hat{c}_l^\dagger, \hat{c}_m^\dagger \right\} = 0$$
$$\left\{ \hat{c}_l, \hat{c}_m^\dagger \right\} = \delta_{l,m}$$

1. Occupation number only 0 and one: nice for computational handling
2. Anti-commute: not so nice to handle in the preparation

## Particle Types

### Introduction of the Physical Problem

- Bosonic operators
  - Commutation relations $\left[\hat{\mathrm{h}}_l, \hat{\mathrm{h}}_m\right] = 0$
- Fermionic operators
  - Anti-commutation relations $\left[\hat{\mathrm{h}}_l^\dagger, \hat{\mathrm{h}}_m^\dagger\right] = 0$
- Hard-core bosonic operators
  - Commutation relations $\left[\hat{\mathrm{h}}_l, \hat{\mathrm{h}}_m^\dagger\right] = \left(1 - 2 \cdot \hat{\mathrm{h}}_m^\dagger \hat{\mathrm{h}}_m\right) \cdot \delta_{l,m}$
  - Occupation number limited to 0 and 1
  - Number operator is idempotent $\left(\hat{\mathrm{h}}_l^\dagger \hat{\mathrm{h}}_l\right)^n = \hat{\mathrm{h}}_l^\dagger \hat{\mathrm{h}}_l \qquad \forall n \in \mathbb{N}$

1. Hard-core bosons: best fo both worlds
2. Like the fermions nice because only occupation number 0 and 1
3. but COMMUTATION relations, because of that we will be able to generate an important mapping
4. Finally: number operator is *idempotent* (power = potenz in english) which we need for one mathematical step later

## Particle Types

### Introduction of the Physical Problem

- Bosonic operators
  - Commutation relations
- Fermionic operators
  - Anti-commutation relations
- Hard-core bosonic operators
  - Commutation relations
  - Occupation number limited to 0 and 1
  - Number operator is idempotent
- Spins (Pauli operators/matrices)
  - Particles from different sites commute
  - Different spin directions on the same site anti-commute
  - They obey angular momentum commutation relations

$$\hat{\sigma}^0 = \mathbb{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\hat{\sigma}^2 = \hat{\sigma}^y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\hat{\sigma}^1 = \hat{\sigma}^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\hat{\sigma}^3 = \hat{\sigma}^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\left\{\hat{\sigma}_l^\alpha, \hat{\sigma}_l^\beta\right\} = 2 \cdot \delta_{\alpha,\beta}$$

$$\left[\hat{\sigma}_l^\alpha, \hat{\sigma}_m^\beta\right] = 2 \cdot i \cdot \varepsilon_{\alpha,\beta,\gamma} \cdot \delta_{l,m} \cdot \hat{\sigma}_l^\gamma$$

1. Final particle type: spins - watch the 1/2 if it is a pauli or spin operator
2. Has also mostly commutative properties (also mainly *angular momentum* relations), but also one anti-commutation relation because of the properties of the Pauli matrices
3. Will be necessary for one mathematical re-formulation

## Particle Types - Mappings

### Introduction of the Physical Problem

- Properties of Pauli matrices
  - Base of the complex 2x2 matrices
  - Can be directly mapped to hard-core bosons

$$\hat{\sigma}_l^x = \hat{h}_l + \hat{h}_l^\dagger$$
$$\hat{\sigma}_l^y = i \cdot \left(\hat{h}_l - \hat{h}_l^\dagger\right)$$
$$\hat{\sigma}_l^z = 2 \cdot \hat{h}_l^\dagger \hat{h}_l - 1$$

$$|\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |0\rangle$$
$$\hat{\sigma}^+ |\downarrow\rangle = |\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |1\rangle = \hat{h}^\dagger |0\rangle$$

1. Pauli matrices are a base for the complex 2x2 matrices
2. Kronecker product of two pauli matrices therefore base for complex 4x4 matrices
3. Simple transformation that can be verified for its commutation relations or derived from two *Jordan-Wigner Transformations*
4. Important to watch the order of base states and keep convention or -1 will be introduced unexpectedly

## Hamiltonian of the system

**Introduction of the Physical Problem**

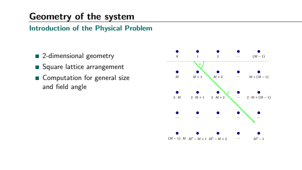
- Hubbard model Hamiltonian
- Constant external electric field
- Two particle types (spin degrees)
  - Different *spins* than for Pauli spin operators

$$\mathcal{H} = \mathcal{H}_0 + \hat{\mathrm{V}}$$

$$\mathcal{H}_0 = U \cdot \sum_{l} \hat{\mathrm{n}}_{l,\uparrow}\hat{\mathrm{n}}_{l,\downarrow} + \sum_{l,\sigma} \underbrace{\left(\vec{E} \cdot \vec{r}_l\right)}_{\varepsilon_l} \hat{\mathrm{n}}_{l,\sigma}$$

$$\hat{\mathrm{V}} = -J \cdot \sum_{\langle l,m\rangle,\sigma} \left(\hat{\mathrm{h}}^{\dagger}_{l,\sigma}\hat{\mathrm{h}}_{m,\sigma} + \hat{\mathrm{h}}^{\dagger}_{m,\sigma}\hat{\mathrm{h}}_{l,\sigma}\right)$$

1. Hubbard model:
   - Double occupation term
   - Two particle degrees of freedom (= spin but not same spin as before: so also written as h and d)
   - Here: electrical field that acts symmetrical on both spin directions
   - Hopping term that is here treated as a perturbation




1. two-dimensional
2. square
3. Electrical field and we want to calculate generally for angle, field strength and number of sites

**Goal of the Calculation**

**Time Evolution**

- Evaluate the time-evolution of the system for various observables
  - Uses calculation in the *Interaction Picture*
  - Introduce operator-less *effective Hamiltonian*

$$\left|\Psi^{\mathrm{S}}(t)\right\rangle = \sum_N e^{\mathcal{H}_{\mathrm{eff}}(N,t)} \Psi_N \left|N\right\rangle$$

1. Goal: evaluate the time-evolution for different observables (effectively)
2. Brute-force calculation is at least NP-hard, definitely exponential and one of the hardest problems to solve computationally generally
3. Strategy: perform a controlled expansion in the perturbation V and re-express the formulas as a dependency on a operator-free "effective Hamiltonian"
   - Can be re-ordered without concern as no operators
   - For the re-formulation we will use the interaction picture
4. Particle-type dependency is stored in this formula in the BASE STATES. Because in this re-formulation, the Heff is a pure number (no operator) and the initial state is a number (not operator dependent at all). But the base-state contains different info for fermions than for bosons, as the operators for hc-bosons commute and so the base-states are uniquely defined. But there are multiple possibilities for teh fermions, because swaps introduce minus signs. (State is not just 0s and 1s, but vacuum that is pre-set with operators and they have an order!!!!!)

## The effective Hamiltonian

Time Evolution

$$\mathcal{H}_{\text{eff}}(N,t) = -iE_0(N)t + \mathcal{H}_N(t)$$

- Constructed from the contributions of the base energy and the perturbation
- Base energy contribution:

$$\begin{aligned} E_0(N) &= \frac{\langle N|\mathcal{H}_0|N\rangle}{\langle N|N\rangle} \\ &= \langle N|U \cdot \sum_l \hat{\mathrm{n}}_{l,\uparrow}\hat{\mathrm{n}}_{l,\downarrow}|N\rangle + \langle N|\sum_{l,\sigma} \varepsilon_l \hat{\mathrm{n}}_{l,\sigma}|N\rangle \\ &= U \cdot \sum_l n_{l,\uparrow}n_{l,\downarrow} + \sum_{l,\sigma} \varepsilon_l n_{l,\sigma} \end{aligned}$$

1. Comprised of two parts:
   - Simple, quickly stated term that depends on the base energy
   - Complicated one that will be expanded in the perturbation

## The effective Hamiltonian

### Time Evolution

- To evaluate the time-evolution of the perturbation

$$\hat{\mathrm{V}} = -J \cdot \sum_{\langle l,m \rangle} \left( \hat{\mathrm{h}}_l^\dagger \hat{\mathrm{h}}_m + \hat{\mathrm{h}}_m^\dagger \hat{\mathrm{h}}_l + \hat{\mathrm{d}}_l^\dagger \hat{\mathrm{d}}_m + \hat{\mathrm{d}}_m^\dagger \hat{\mathrm{d}}_l \right)$$

- Solve the equation of motion for the ladder operators

$$\begin{aligned}
\hat{\mathrm{h}}_m^{\dagger\mathrm{I}}(t) &= e^{i \cdot \varepsilon_m \cdot t} \left( 1 + \left( e^{i \cdot U \cdot t} - 1 \right) \hat{\mathrm{d}}_m^{\dagger\mathrm{S}} \hat{\mathrm{d}}_m^{\mathrm{S}} \right) \hat{\mathrm{h}}_m^{\dagger\mathrm{S}} \\
\hat{\mathrm{h}}_m^{\mathrm{I}}(t) &= e^{-i \cdot \varepsilon_m \cdot t} \left( 1 + \left( e^{-i \cdot U \cdot t} - 1 \right) \hat{\mathrm{d}}_m^{\dagger\mathrm{S}} \hat{\mathrm{d}}_m^{\mathrm{S}} \right) \hat{\mathrm{h}}_m^{\mathrm{S}} \\
\hat{\mathrm{d}}_m^{\dagger\mathrm{I}}(t) &= e^{i \cdot \varepsilon_m \cdot t} \left( 1 + \left( e^{i \cdot U \cdot t} - 1 \right) \hat{\mathrm{h}}_m^{\dagger\mathrm{S}} \hat{\mathrm{h}}_m^{\mathrm{S}} \right) \hat{\mathrm{d}}_m^{\dagger\mathrm{S}} \\
\hat{\mathrm{d}}_m^{\mathrm{I}}(t) &= e^{-i \cdot \varepsilon_m \cdot t} \left( 1 + \left( e^{-i \cdot U \cdot t} - 1 \right) \hat{\mathrm{h}}_m^{\dagger\mathrm{S}} \hat{\mathrm{h}}_m^{\mathrm{S}} \right) \hat{\mathrm{d}}_m^{\mathrm{S}}
\end{aligned}$$

1. Need: V-Operator in the Interaction picture
2. For that: first solve the equations of motion for the ladder operators and the plug in
3. Solving the equations of motions in the interaction picture requires their number operator being *idempotent*

## The effective Hamiltonian

### Time Evolution

- Insert and reorder the operators

$$
\begin{aligned}
\hat{\mathrm{V}}^{\mathrm{I}}(t) &= \left\{\hat{\mathrm{V}}^{\mathrm{S}}\right\}^{\mathrm{I}}(t) = -J \cdot \sum_{[l,m]} \left\{\left(\hat{\mathrm{h}}_l^{\dagger\mathrm{S}}\hat{\mathrm{h}}_m^{\mathrm{S}} + \hat{\mathrm{d}}_l^{\dagger\mathrm{S}}\hat{\mathrm{d}}_m^{\mathrm{S}}\right)\right\}^{\mathrm{I}}(t) \\
&= -J \cdot \sum_{[l,m]} \left(\hat{\mathrm{h}}_l^{\dagger\mathrm{I}}(t)\hat{\mathrm{h}}_m^{\mathrm{I}}(t) + \hat{\mathrm{d}}_l^{\dagger\mathrm{I}}(t)\hat{\mathrm{d}}_m^{\mathrm{I}}(t)\right) \\
&= -J \cdot \sum_{[l,m]} \left[\Lambda_{\mathrm{A}}(l,m,t) \cdot \hat{\mathrm{F}}_{\mathrm{A}}(l,m) + \Lambda_{\mathrm{B}}(l,m,t) \cdot \hat{\mathrm{F}}_{\mathrm{B}}(l,m) + \Lambda_{\mathrm{C}}(l,m,t) \cdot \hat{\mathrm{F}}_{\mathrm{C}}(l,m)\right]
\end{aligned}
$$

$$
\begin{aligned}
\Lambda_{\mathrm{A}}(l,m,t) &= e^{i\cdot(\varepsilon_l-\varepsilon_m)\cdot t} & \hat{\mathrm{F}}_{\mathrm{A}}(l,m) &= \sum_{\sigma\in\{\uparrow,\downarrow\}} \hat{\mathrm{h}}_{l,\sigma}^{\dagger\mathrm{S}}\hat{\mathrm{h}}_{m,\sigma}^{\mathrm{S}}\left(1 + 2\cdot\hat{\mathrm{n}}_{l,\overline{\sigma}}^{\mathrm{S}}\hat{\mathrm{n}}_{m,\overline{\sigma}}^{\mathrm{S}} - \hat{\mathrm{n}}_{l,\overline{\sigma}}^{\mathrm{S}} - \hat{\mathrm{n}}_{m,\overline{\sigma}}^{\mathrm{S}}\right) \\
\Lambda_{\mathrm{B}}(l,m,t) &= e^{i\cdot(\varepsilon_l-\varepsilon_m+U)\cdot t} & \hat{\mathrm{F}}_{\mathrm{B}}(l,m) &= \sum_{\sigma\in\{\uparrow,\downarrow\}} \hat{\mathrm{h}}_{l,\sigma}^{\dagger\mathrm{S}}\hat{\mathrm{h}}_{m,\sigma}^{\mathrm{S}}\left(\hat{\mathrm{n}}_{l,\overline{\sigma}}^{\mathrm{S}} - \hat{\mathrm{n}}_{l,\overline{\sigma}}^{\mathrm{S}}\hat{\mathrm{n}}_{m,\overline{\sigma}}^{\mathrm{S}}\right) \\
\Lambda_{\mathrm{C}}(l,m,t) &= e^{i\cdot(\varepsilon_l-\varepsilon_m-U)\cdot t} & \hat{\mathrm{F}}_{\mathrm{C}}(l,m) &= \sum_{\sigma\in\{\uparrow,\downarrow\}} \hat{\mathrm{h}}_{l,\sigma}^{\dagger\mathrm{S}}\hat{\mathrm{h}}_{m,\sigma}^{\mathrm{S}}\left(\hat{\mathrm{n}}_{m,\overline{\sigma}}^{\mathrm{S}} - \hat{\mathrm{n}}_{m,\overline{\sigma}}^{\mathrm{S}}\hat{\mathrm{n}}_{l,\overline{\sigma}}^{\mathrm{S}}\right)
\end{aligned}
$$

1. Inserting yields TWO kinds of terms
   - time-dependent analytical pre-factors (or their integral as one sees later)
   - dressed hopping terms, basically only depending on the occupation of the state N

## The effective Hamiltonian

### Time Evolution

- Evaluate the contribution to the effective Hamiltonian
  - Requires previously calculated value of the V-operator in the Interaction Picture
  - Controllable *cumulant expansion*

$$
\begin{aligned}
\mathcal{H}_N(t) &= \sum_{v=1}^{\infty} \frac{(-i)^v}{v!} \int_0^t \mathrm{d}t_1 \int_0^t \mathrm{d}t_2 \cdots \int_0^t \mathrm{d}t_v \left\langle \mathbb{T}\, \hat{\mathrm{V}}^{\mathrm{I}}(t_1) \hat{\mathrm{V}}^{\mathrm{I}}(t_2) \cdots \hat{\mathrm{V}}^{\mathrm{I}}(t_v) \right\rangle_{c(N)} \\
&= -i \int_0^t \mathrm{d}t_1 \frac{\left\langle N \middle| \hat{\mathrm{V}}^{\mathrm{I}}(t_1) \middle| \Psi^S \right\rangle}{\langle N | \Psi^S \rangle} \\
&- \frac{1}{2} \int_0^t \mathrm{d}t_1 \int_0^t \mathrm{d}t_2 \left( \frac{\left\langle N \middle| \mathbb{T} \hat{\mathrm{V}}^{\mathrm{I}}(t_1) \hat{\mathrm{V}}^{\mathrm{I}}(t_2) \middle| \Psi^S \right\rangle}{\langle N | \Psi^S \rangle} - \frac{\left\langle N \middle| \hat{\mathrm{V}}^{\mathrm{I}}(t_1) \middle| \Psi^S \right\rangle \cdot \left\langle N \middle| \hat{\mathrm{V}}^{\mathrm{I}}(t_2) \middle| \Psi^S \right\rangle}{\langle N | \Psi^S \rangle^2} \right) \\
&+ \cdots
\end{aligned}
$$

1. Generally any expansion possible, but here we expand the *argument of the exponential* - cumulant
2. first order easily derivable with the info from the previous slide (integrate)
3. second and higher orders very complicated occupation terms and the pre-factors require quite a lot of edge-cases to integrate with the time-ordering operator
   - Hint onto the thesis for this exact derivation

## Handling of Observables

### Time Evolution

- This allows for general evaluation of expectation values
  - Requires a probability for the state
  - Requires a local observable

$$\frac{\left\langle \Psi^S(t) \middle| \hat{\mathcal{O}} \middle| \Psi^S(t) \right\rangle}{\langle \Psi^S(t) | \Psi^S(t) \rangle} =$$

$$\sum_N \mathrm{P}(N,t) \underbrace{\sum_K \langle N | \hat{\mathcal{O}} | K \rangle \, e^{\mathcal{H}_{\mathrm{eff}}(K,t) - \mathcal{H}_{\mathrm{eff}}(N,t)} \frac{\Psi_K}{\Psi_N}}_{\hat{\mathcal{O}}_{\mathrm{loc}}(N,t)} = \sum_N \mathrm{P}(N,t) \hat{\mathcal{O}}_{\mathrm{loc}}(N,t)$$

1. Why? General evaluation of an expectation value for an observable.
   - Probability for state
   - Local observable (can be swapped for many different ones)
2. Does ONLY require difference of effective Hamiltonian for evaluation (notice for later)

## Simple Observables

### Time Evolution

- Local observable for double occupation measurement
  - Observables generally are very sparse matrices
  - Reduces to pure occupation measurement

$$\hat{\mathcal{O}}_{\text{loc}}(N,t) = \sum_K \langle N|\hat{\mathcal{O}}_{\text{do}}(l)|K\rangle\, e^{\mathcal{H}_{\text{eff}}(K,t)-\mathcal{H}_{\text{eff}}(N,t)} \frac{\Psi_K}{\Psi_N} = n_{l,\uparrow} \cdot n_{l,\downarrow}$$

- Local observable for particle current measurement
  - Requires evaluation of the difference of two effective Hamiltonians

$$\hat{\mathcal{O}}_{\text{loc}}(N,t) = i \cdot \left[n_{m,\sigma} \cdot (1-n_{l,\sigma}) - n_{l,\sigma} \cdot (1-n_{m,\sigma})\right] \cdot \frac{\Psi_{\tilde{N}}}{\Psi_N} \cdot e^{\mathcal{H}_{\text{eff}}(\tilde{N},t)-\mathcal{H}_{\text{eff}}(N,t)}$$

1. Complex valued observable, make sure this cancels in the case of approximations
2. *SHOULD MENTION:* Energy and variance (most expensive ones, are assumed to keep constant, can be used to gauge the quality of the expansion)

## Access to Density-Matrics

### Time Evolution

- Non-classical (*quantum*) measurements depend on the (reduced) density matrix
  - Access to *purity*, *concurrence* and other entanglement measures/monotones
- Direct calculation not possible

$$\rho_A = \mathrm{Tr}_B\left(\rho_{AB}\right) = \sum_{k=0}^{M} \left(\mathbb{1}_A \otimes \langle \mathcal{X}_k|\right) \rho_{AB} \left(\mathbb{1}_A \otimes |\mathcal{X}_k\rangle\right)$$

- Use Pauli matrices to expand the complex 4x4 matrix

$$\rho_A(t) = \rho_{l,\sigma,m,\mu}(t) = \frac{1}{4} \sum_{\alpha,\beta\in\{0,x,y,z\}} \left\langle \hat{\sigma}^{\alpha}_{l,\sigma}\hat{\sigma}^{\beta}_{m,\mu} \right\rangle_{(t)} \left(\hat{\sigma}^{\alpha} \otimes \hat{\sigma}^{\beta}\right)$$

1. Quite interesting, because of this extra mentioned the Pauli operators
2. No access to the full density matrix, can not trace out, because too many states to trace out
3. BUT: as the complex 4x4 matrix can be written in the base of the kronecker product of two pauli matrices, can be expressed (because the pauli operators can be translated into hard-core bosonic ones again and for these we have the general fromula for expectation values)

## Access to Density-Matrics

### Time Evolution

$$\hat{\mathcal{O}} = \hat{\sigma}^{x}_{l,\sigma}\hat{\sigma}^{0}_{m,\mu} : \quad \hat{\mathcal{O}}_{\text{loc}}(N,t) = \frac{\Psi_{\tilde{N}}}{\Psi_{N}} \cdot e^{\mathcal{H}_{\text{eff}}(\tilde{N},t)-\mathcal{H}_{\text{eff}}(N,t)}$$

$$\hat{\mathcal{O}} = \hat{\sigma}^{y}_{l,\sigma}\hat{\sigma}^{0}_{m,\mu} : \quad \hat{\mathcal{O}}_{\text{loc}}(N,t) = \frac{\Psi_{\tilde{N}}}{\Psi_{N}} \cdot e^{\mathcal{H}_{\text{eff}}(\tilde{N},t)-\mathcal{H}_{\text{eff}}(N,t)} \cdot i \cdot (1 - 2 \cdot n_{l,\sigma})$$

$$\hat{\mathcal{O}} = \hat{\sigma}^{z}_{l,\sigma}\hat{\sigma}^{0}_{m,\mu} : \quad \hat{\mathcal{O}}_{\text{loc}}(N,t) = \frac{\Psi_{\tilde{N}}}{\Psi_{N}} \cdot e^{\mathcal{H}_{\text{eff}}(\tilde{N},t)-\mathcal{H}_{\text{eff}}(N,t)} \cdot (2 \cdot n_{l,\sigma} - 1)$$

$$\hat{\mathcal{O}} = \hat{\sigma}^{x}_{l,\sigma}\hat{\sigma}^{x}_{m,\mu} : \quad \hat{\mathcal{O}}_{\text{loc}}(N,t) = \frac{\Psi_{\tilde{\tilde{N}}}}{\Psi_{N}} \cdot e^{\mathcal{H}_{\text{eff}}(\tilde{\tilde{N}},t)-\mathcal{H}_{\text{eff}}(N,t)}$$

$$\hat{\mathcal{O}} = \hat{\sigma}^{y}_{l,\sigma}\hat{\sigma}^{y}_{m,\mu} : \quad \hat{\mathcal{O}}_{\text{loc}}(N,t) = \frac{\Psi_{\tilde{\tilde{N}}}}{\Psi_{N}} \cdot e^{\mathcal{H}_{\text{eff}}(\tilde{\tilde{N}},t)-\mathcal{H}_{\text{eff}}(N,t)} \cdot (2 \cdot n_{l,\sigma} - 1) \cdot (1 - 2 \cdot n_{m,\mu})$$

$$\hat{\mathcal{O}} = \hat{\sigma}^{x}_{l,\sigma}\hat{\sigma}^{y}_{m,\mu} : \quad \hat{\mathcal{O}}_{\text{loc}}(N,t) = \frac{\Psi_{\tilde{\tilde{N}}}}{\Psi_{N}} \cdot e^{\mathcal{H}_{\text{eff}}(\tilde{\tilde{N}},t)-\mathcal{H}_{\text{eff}}(N,t)} \cdot i \cdot (1 - 2 \cdot n_{m,\mu})$$

$$\hat{\mathcal{O}} = \hat{\sigma}^{y}_{l,\sigma}\hat{\sigma}^{x}_{m,\mu} : \quad \hat{\mathcal{O}}_{\text{loc}}(N,t) = \frac{\Psi_{\tilde{\tilde{N}}}}{\Psi_{N}} \cdot e^{\mathcal{H}_{\text{eff}}(\tilde{\tilde{N}},t)-\mathcal{H}_{\text{eff}}(N,t)} \cdot i \cdot (1 - 2 \cdot n_{l,\sigma})$$

1. Sixteen terms (with missing cases trivially derived from the ones that are given)
2. Again all depend on the difference of effective hamiltonians between two states ith local modifications

## Probability Calculations

### Sampling & Simplifications

- Full probability still requires normalization
- Switch to the Metropolis *Hastings* algorithm
  - Non-exponential number of samples
  - Depends only on a transition probability
  - Does not need normalization because it cancels

$$\alpha = \frac{\mathrm{P}(\widetilde{N},t)}{\mathrm{P}(N,t)} = \frac{f(\widetilde{N},t)}{f(N,t)} = \frac{\left|e^{\mathcal{H}_{\text{eff}}(\widetilde{N},t)}\right|^2 |\Psi_{\widetilde{N}}|^2}{\left|e^{\mathcal{H}_{\text{eff}}(N,t)}\right|^2 |\Psi_N|^2}$$
$$= \frac{|\Psi_{\widetilde{N}}|^2}{|\Psi_N|^2} \frac{e^{\Re(\mathcal{H}_{\text{eff}}(\widetilde{N},t)) + i\Im(\mathcal{H}_{\text{eff}}(\widetilde{N},t)) + \Re(\mathcal{H}_{\text{eff}}(\widetilde{N},t)) - i\Im(\mathcal{H}_{\text{eff}}(\widetilde{N},t))}}{e^{\Re(\mathcal{H}_{\text{eff}}(N,t)) + i\Im(\mathcal{H}_{\text{eff}}(N,t)) + \Re(\mathcal{H}_{\text{eff}}(N,t)) - i\Im(\mathcal{H}_{\text{eff}}(N,t))}}$$
$$= \frac{|\Psi_{\widetilde{N}}|^2}{|\Psi_N|^2} e^{2\cdot\Re(\mathcal{H}_{\text{eff}}(\widetilde{N},t)) - 2\cdot\Re(\mathcal{H}_{\text{eff}}(N,t))}$$
$$= \frac{|\Psi_{\widetilde{N}}|^2}{|\Psi_N|^2} e^{2\cdot\Re(\mathcal{H}_{\text{eff}}(\widetilde{N},t) - \mathcal{H}_{\text{eff}}(N,t))}$$

$$\frac{\left\langle \Psi^{\mathrm{S}}(t) \middle| \hat{\mathcal{O}} \middle| \Psi^{\mathrm{S}}(t) \right\rangle}{\langle \Psi^{\mathrm{S}}(t) | \Psi^{\mathrm{S}}(t) \rangle} = \sum_N \mathrm{P}(N,t) \cdot \hat{\mathcal{O}}_{\text{loc}}(N,t) \approx \frac{1}{|\{N\}_{\text{MC}}|} \sum_{\{N\}_{\text{MC}}} \hat{\mathcal{O}}_{\text{loc}}(N,t)$$

1. Normalization requires knowing exponential amount of states and we need to sample an exponential amount to get all
2. switch to METROPOLIS HASTINGS algorithm
   - only fixed number of samples required (depends on the desired precision)
   - No normalization needed, because it cancels in the division

## Analytical Simplifications

### Sampling & Simplifications

- Choose a helpful initial state

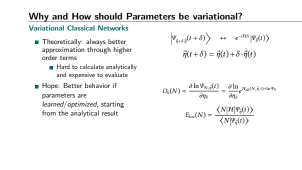
$$\Psi_N = \frac{1}{\sqrt{\#(\text{states})}} = \frac{1}{\sqrt{2^{\#(\text{sites})\cdot 2}}} = \frac{1}{2^{\#(\text{sites})}}$$

$$\left|\Psi^S(t=0)\right\rangle = \bigotimes_{l=1}^{\#(\text{states})} \frac{1}{2}\left(1 + \hat{\mathrm{h}}^{\dagger S}_{l,\uparrow} + \hat{\mathrm{h}}^{\dagger S}_{l,\downarrow} + \hat{\mathrm{h}}^{\dagger S}_{l,\uparrow}\hat{\mathrm{h}}^{\dagger S}_{l,\downarrow}\right)|0\rangle$$

- Now *almost all* terms cancel in *differences* of the effective Hamiltonian
  - Exemplary for single flip on base energy term

$$\begin{aligned} E_0(N) - E_0(\widetilde{N}) &= U\sum_l n_{l,\downarrow}n_{l,\uparrow} - U\sum_l \widetilde{n}_{l,\downarrow}\widetilde{n}_{l,\uparrow} + \sum_{l,\sigma}\varepsilon_l n_{l,\sigma} - \sum_{l,\sigma}\varepsilon_l \widetilde{n}_{l,\sigma} \\ &= \varepsilon_i\left(2n_{i,\sigma_i} - 1\right) + U\cdot\begin{cases}\sigma_i = \uparrow: & n_{i,\downarrow}(2n_{i,\uparrow} - 1) \\ \sigma_i = \downarrow: & n_{i,\uparrow}(2n_{i,\downarrow} - 1)\end{cases} \end{aligned}$$

1. Choose the helpful *initial state* with all base states having the same probability
2. because of this the divisions of Psi-N over Psi-K everywhere give 1 and many terms cancel (outside of the local interaction ragne of the modificaiton)




1. Argument:
   - Could go on performing the controlled expansion for all perturbation-strengths and get nice result
   - But higher orders complicated and expensive
   - Try to get more from the lower orders, by using Variational parameters (optimized by the problem, we hope they can incorperate influence from higher order terms without them being present)
2. Time dependent variational principle (TDVP)
   - Find a parametrized *wavefunction*
   - Update the parameters so that a tiny nudge into the direction of the *time DERIVATIVE* of the parameter vector has the same effect as a time evolution with the full hamiltonian
   - The first order numerical integration (could do higher orders, but for the concept here)
3. O: Variational derivative E: local energy (we had already earlier)

## Why and How should Parameters be variational?

### Variational Classical Networks

- Theoretically: always better approximation through higher order terms
  - Hard to calculate analytically and expensive to evaluate
- Hope: Better behavior if parameters are *learned/optimized*, starting from the analytical result

$$\left|\Psi_{\vec{\eta}+\delta\cdot\dot{\vec{\eta}}}(t+\delta)\right\rangle \quad \leftrightarrow \quad e^{-i\mathcal{H}\delta}\left|\Psi_{\vec{\eta}}(t)\right\rangle$$

$$\vec{\eta}(t+\delta) = \vec{\eta}(t) + \delta\cdot\dot{\vec{\eta}}(t)$$

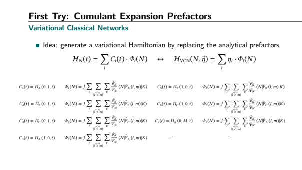
$$\overleftrightarrow{S}_{k,k'} = \left\langle O_k^* O_{k'}\right\rangle_{(\vec{\eta})} - \left\langle O_k^*\right\rangle_{(\vec{\eta})}\cdot\left\langle O_{k'}\right\rangle_{(\vec{\eta})}$$

$$\vec{F}_k = \left\langle E_{\mathrm{loc}} O_k^*\right\rangle_{(\vec{\eta})} - \left\langle E_{\mathrm{loc}}\right\rangle_{(\vec{\eta})}\cdot\left\langle O_k^*\right\rangle_{(\vec{\eta})}$$

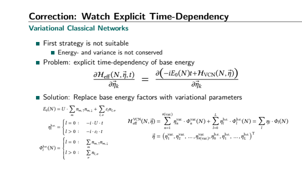
$$\sum_{k'} \overleftrightarrow{S}_{k,k'}\dot{\vec{\eta}}_{k'} = -i\vec{F}_k$$

$$\dot{\vec{\eta}} = -i\overleftrightarrow{S}^{-1}\vec{F}$$

1. Pseudo inversion
2. S: covariance matrix
3. F: TDVP force
4. *time DERIVATIVE*




1. Idea: replace prefactors that depend on the time and keep the shape of teh terms that depend on the occupation of the state
2. Verification: as the TDVP is also derived from an *action principle* that must mean as per noethers theorem that the energy MUST be conserved by a correct TDVP step




1. Fails because not taken the explicit time-dependency into account as it was *differentiated away* wrongfully
2. As a solution: introduce variational base-energy factors - which is a successful strategy

**Implementation**

Implementation & Experiments

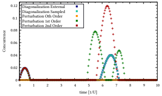
- Too much code to properly show
  - Look at repository for overview
- Testing: validation code to check the correct implementation
  - Independent implementations for same functionality
  - Sanity checks for state of data and program
- Optimization: runtime comparison for different code styles
  - For generated code fast execution over readability

1. While nothing is looked at in detail maybe interesting for some
2. Testing: validation and assertion based programming
3. Optimization: code style runtime experiments

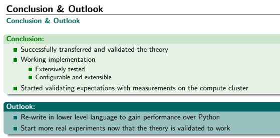
## Numerical Experiments

**Implementation & Experiments**

- Sadly no presentation possible in the small time allocated for the presentation
- The plots are in the presentation in the supplementary material




1. See supplementary material for more plots and stuff



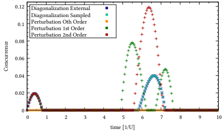

1. Conclusion:
   - Managed to Solve all problems
   - Provide heavily *tested* and *extensible* library of code AND theoretical summary
   - Started to validate expectations with computations on the cluster
2. Future plans:
   - re-write in performant language after testing is successful
   - Go through the parameters to start investigating the behavior of the systems

Thank you for your kind attention

1. Thank you for your kind attention
2. All tools and other resources are referenced in the presentation
3. You can also find everything on my Github

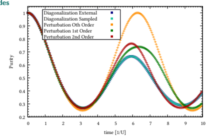
**Exact Comparison: Concurrence**

Extra slides




1. Concurrence comparison of small system with exact calculations and comparison thereof
2. See extreme boost of second order
3. NOTICE: how the External calculation is done COMPLETELY on density matrices and diagonalization, so prooves we can sample the reduced density matrix out

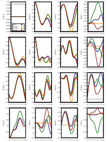
Exact Comparison: Purity
Extra slides




1. Purity measurements for the graph of the measurement from one page before

**Exact Comparison: Pauli Measurements**

Extra slides



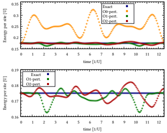
1. Portrait mode format, because extracted from the thesis.
2. You can see what a problem the z-contributions are

2025-03-12

Master Colloquium Presentation
└─Extra slides
└─First Orders: Energy Comparison

**First Orders: Energy Comparison**

Extra slides




1. Second picture zoomed in

2025-03-12

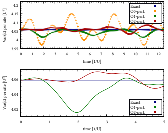
Master Colloquium Presentation

Extra slides

First Orders: Variance Comparison

**First Orders: Variance Comparison**

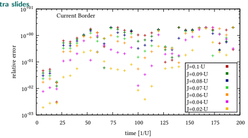
Extra slides




1. Second picture zoomed in

**J-Sweep: Current Border**

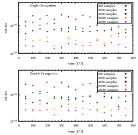
Extra slides




1. Logarithmic plot
2. Measurements show the trend of the convergence of the relative error for small perturbations
3. Depends on the observable though, still

## Monte Carlo Convergence: Occupation

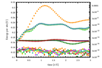
Extra slides

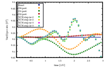



1. Well as we can see, the appriximation converges as the variance gos to zero

**VCN Failiure: Non-Explicit Parametrization**



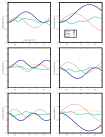




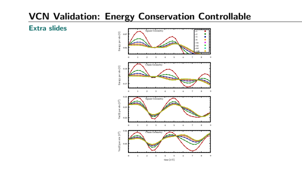
1. Complex part now properly vanishes (was long a problem)
2. See how the vcn goes like the base energy curve only in the beginning (suspiciously)

**VCN Failiure: Non-Explicit Parametrization**

**Extra slides**



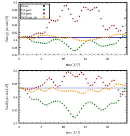
1. See complex part jetting off to infinity because it doesn't knwo how to compensate




1. Still no perfect conservation for infinite steps, but this is probably due to first order integration only

**VCN: Small Square System**

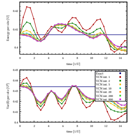
Extra slides




1. Notice how VCN now is MUCH better

**VCN: Small Square System**

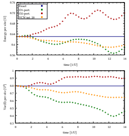
**Extra slides**




1. Step size dependent version of previous slide

## VCN: Bigger Square System

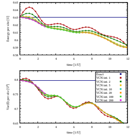
Extra slides




1. Now not a guarantee for immediately better results, but very promising

## VCN: Bigger Square System

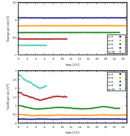
Extra slides




1. Here one can see very quickly more steps doesn't only make it better

**VCN End to End Test: System Size Dependency**

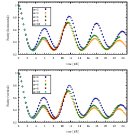
Extra slides



1. This is the energy for validation

**VCN End to End Test: System Size Dependency**

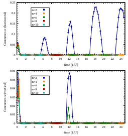
Extra slides




1. This is the purity
2. Does not seem to have a huge variation

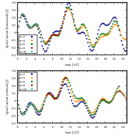
**VCN End to End Test: System Size Dependency**

Extra slides




1. This is the concurrence
2. Poorly resolved, this should be looked into possibly in the future

VCN End to End Test: System Size Dependency
Extra slides



1. This is the current
2. Most obvious effect, but still extremely stable over a order of magnitude of sites
3. See assymmetric behavior, because electrical field is set to not be symmetric